

CHINGGIS KHAAN INTL, Mongolia

WMO: 442910

Lat: 47.843N

Lon: 106.767E

Elev: 1330

StdP: 86.33

Time Zone: 8.00 (E08)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.8	-34.9	-40.3	0.1	-36.7	-38.4	0.1	-34.8	8.3	-16.9	7.3	-19.2	1.0	180	0.713

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	30.5	15.3	28.4	14.7	26.6	14.2	17.7	25.7	16.8	24.2	16.0	23.2	4.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.2	12.7	20.0	14.3	11.9	19.4	13.3	11.2	18.7	55.5	25.6	52.5	24.0	49.8	23.2	21.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max					
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
10.8	9.4	8.2	DB	-39.3	33.9	2.0	3.0	-40.8	36.1	-41.9	37.9	-43.0	39.6	-44.5	41.8	
			WB	-39.2	19.3	2.2	1.2	-40.7	20.1	-42.0	20.8	-43.3	21.5	-44.9	22.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.3	-24.9	-20.4	-7.7	2.4	9.0	15.9	18.6	16.0	9.5	-0.2	-13.4	-22.0
	DBStd	16.04	4.82	5.64	7.03	5.28	5.24	4.30	3.34	3.88	4.68	4.73	6.97	4.88
	HDD10.0	4875	1082	852	548	231	82	4	0	2	63	316	703	992
	HDD18.3	7264	1340	1086	806	477	291	95	36	92	265	573	953	1251
	CDD10.0	742	0	0	0	4	52	182	266	189	49	0	0	0
	CDD18.3	91	0	0	0	0	3	23	43	21	1	0	0	0
	CDH23.3	1369	0	0	0	1	101	388	537	301	43	0	0	0
	CDH26.7	397	0	0	0	0	23	114	166	88	6	0	0	0
Wind	WSAvg	3.0	1.6	2.1	3.0	4.1	4.6	4.0	3.5	3.2	3.3	2.9	2.1	1.7
Precipitation	PrecAvg	249	2	2	3	7	14	52	70	61	24	7	3	2
	PrecMax	400	6	6	7	32	41	237	197	135	66	26	12	11
	PrecMin	59	0	0	0	0	0	7	22	4	1	0	0	0
	PrecStd	90	1	2	2	9	13	44	40	38	17	6	3	3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-8.3	-0.5	13.2	22.6	30.1	33.0	33.4	32.3	27.3	17.9	6.5	-7.1
		MCWB	-10.1	-4.7	3.9	8.3	12.5	15.5	16.8	16.5	12.9	6.8	0.8	-8.7
	2%	DB	-11.4	-4.1	9.3	19.0	26.3	29.7	30.5	29.0	24.4	14.3	3.9	-10.0
		MCWB	-12.8	-6.8	1.8	7.0	10.6	14.1	16.5	14.8	12.2	5.6	-0.5	-11.2
	5%	DB	-14.3	-7.2	6.4	16.1	23.5	27.5	28.3	26.8	21.7	11.5	0.5	-12.0
		MCWB	-15.2	-9.1	0.3	5.6	9.6	13.4	15.9	15.0	11.1	4.0	-2.7	-12.9
	10%	DB	-16.7	-10.1	3.4	13.5	20.7	25.4	26.3	24.3	19.4	8.9	-2.1	-14.0
		MCWB	-17.4	-11.5	-1.2	4.3	8.7	12.9	15.4	14.6	10.3	2.8	-4.4	-14.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-10.0	-4.4	4.3	9.1	12.5	17.4	19.4	18.3	14.5	7.3	1.1	-8.6
		MCDB	-8.2	-0.5	12.4	20.7	28.1	26.5	26.4	27.2	23.3	16.3	5.8	-7.1
	2%	WB	-12.8	-6.8	2.1	7.1	11.1	15.8	18.2	17.2	13.3	5.9	-0.5	-11.1
		MCDB	-11.4	-4.2	8.7	17.9	23.9	24.9	26.2	24.5	21.6	13.8	3.4	-10.1
	5%	WB	-15.2	-9.1	0.6	6.0	10.1	14.9	17.3	16.2	12.3	4.4	-2.6	-12.9
		MCDB	-14.2	-7.2	6.1	15.3	22.9	23.2	25.1	23.0	20.1	10.7	0.3	-12.0
	10%	WB	-17.4	-11.5	-1.1	4.6	8.9	14.1	16.6	15.4	11.0	3.1	-4.5	-14.8
		MCDB	-16.7	-10.0	3.5	12.9	19.9	22.1	24.0	21.9	18.2	8.4	-2.2	-14.1
Mean Daily Temperature Range	5% DB	MDBR	13.1	15.8	15.6	15.5	16.0	13.6	12.4	13.0	15.3	14.4	13.2	12.0
		MCDBR	14.7	17.9	18.0	21.1	21.3	17.3	15.9	17.4	19.4	19.2	15.4	13.4
		MCWBR	13.5	15.5	11.2	11.4	9.8	6.3	5.6	6.4	8.7	11.3	11.4	12.3
	5% WB	MCDBR	14.6	17.8	16.9	19.4	19.1	12.8	13.0	12.8	16.3	17.6	14.9	13.4
		MCWBR	13.4	15.4	10.6	10.6	8.9	5.0	5.4	5.3	7.7	10.4	11.1	12.3
Clear-Sky Solar Irradiance	taub		0.240	0.263	0.303	0.349	0.380	0.390	0.375	0.356	0.311	0.283	0.258	0.242
	taud		2.337	2.293	2.269	2.236	2.191	2.265	2.358	2.388	2.454	2.470	2.395	2.330
	Ebn at Noon		860	907	916	900	881	872	881	885	901	877	828	806
	Edh at Noon		69	91	111	128	139	131	117	109	92	76	65	61
All-Sky Solar Radiation	RadAvg		1.94	3.09	4.52	5.69	6.55	6.51	6.14	5.46	4.59	3.17	2.01	1.53
	RadStd		0.11	0.19	0.18	0.24	0.46	0.36	0.34	0.36	0.25	0.16	0.11	0.10

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page